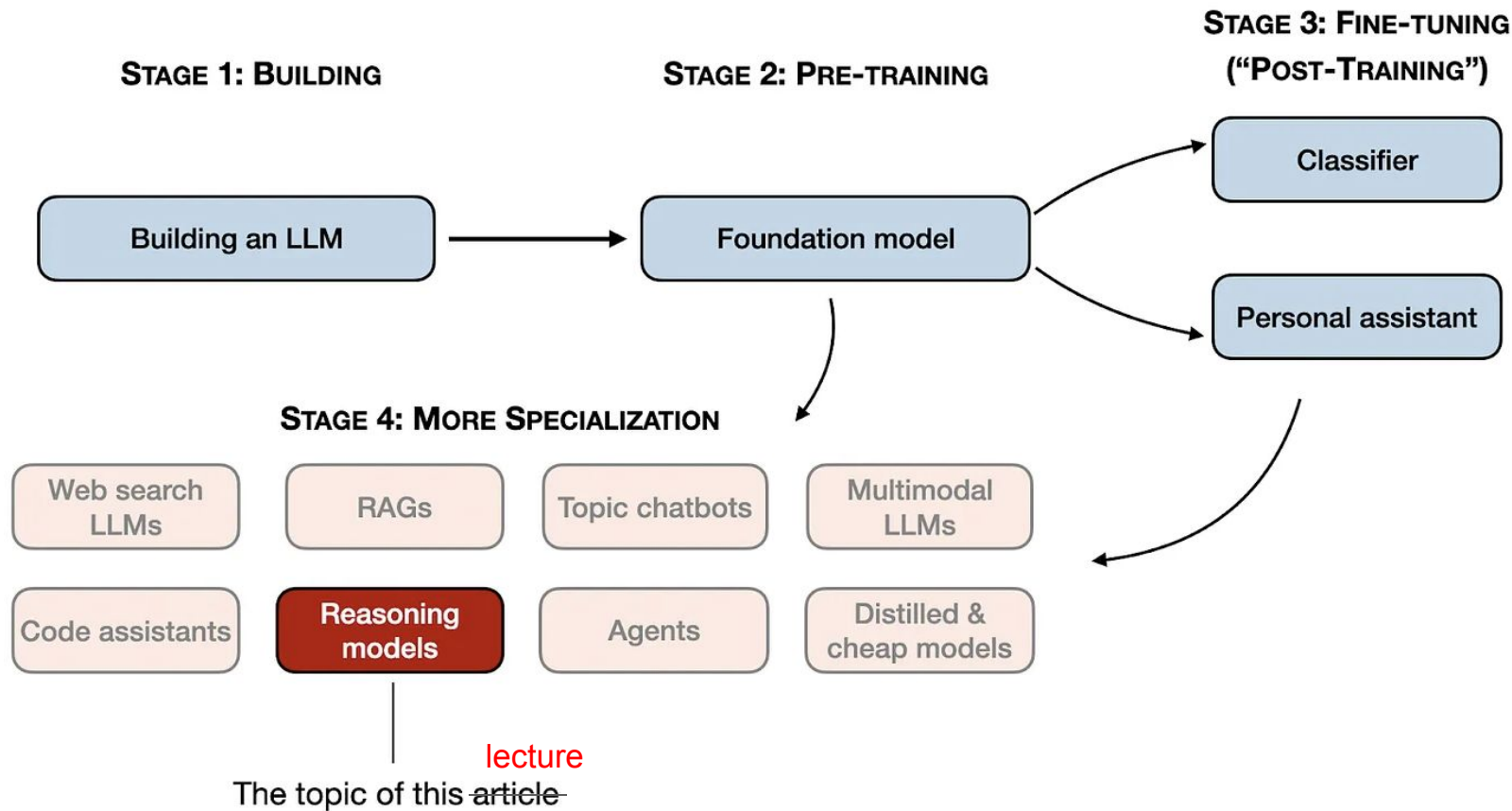
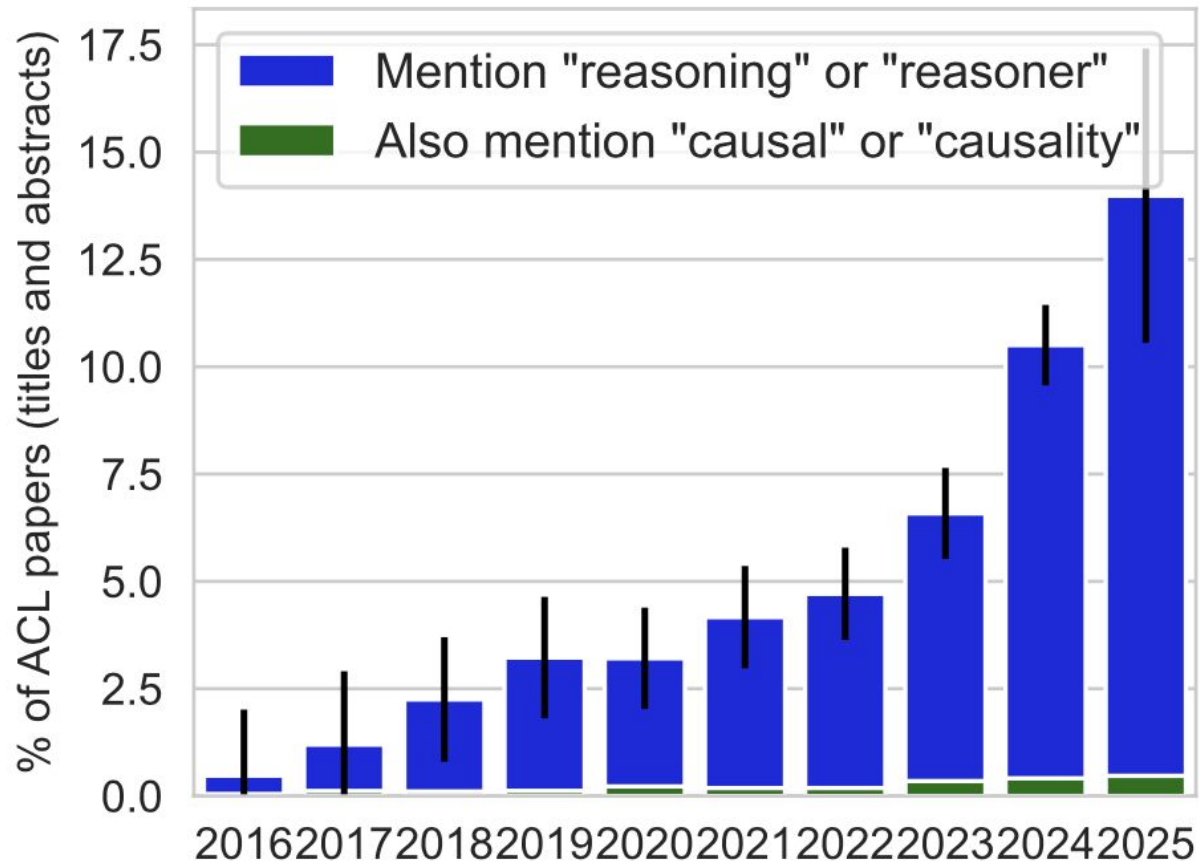


Prompt Engineering and Reasoning in LLMs

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Rodrygo Santos



Stages 1-3 are the common steps to developing LLMs. Stage 4 specializes LLMs for specific use cases.



Causality can systematically address the monsters under the bench(marks), <https://arxiv.org/abs/2502.05085>

Today

Prompt engineering

Reasoning in LLMs

Prompt Engineering

Objectives

Present an introduction to prompt engineering

Present an overview of the latest prompting techniques

Provide demonstrations and exercises to practice different prompting techniques

Agenda

Introduction to Prompt Engineering

Advanced Techniques for Prompt Engineering

Applications & Tools

Conclusion & Future Directions

Part 1

Introduction to Prompt Engineering

What are prompts?

- **Prompts** involve instructions and context passed to a language model to achieve a desired task
- **Prompt engineering** is the practice of developing and optimizing prompts to efficiently use language models (LMs) for a variety of applications
- Prompt engineering is a useful skill for AI engineers and researchers to improve and efficiently use language models

What is prompt engineering?

Prompt engineering is a process of creating a set of prompts, or questions, that are used to guide the user toward a desired outcome. It is an effective tool for designers to create user experiences that are easy to use and intuitive. This method is often used in interactive design and software development, as it allows users to easily understand how to interact with a system or product..

Why Prompt Engineering?

- Why learn prompt engineering?
 - Important for research, discoveries, and advancement
 - Helps to test and evaluate the limitations of LLMs
 - Enables all kinds of innovative applications on top of LLMs

ANTHROPIC

Prompt Engineer and Librarian

SAN FRANCISCO, CA / PRODUCT / FULL-TIME / HYBRID

Anthropic's mission is to create reliable, interpretable, and steerable AI systems. We want AI to be safe for our customers and for society as a whole.

Anthropic's AI technology is amongst the most capable and safe in the world. However, large language models are a new type of intelligence, and the art of instructing them in a way that delivers the best results is still in its infancy – it's a hybrid between programming, instructing, and teaching. You will figure out the best methods of prompting our AI to accomplish a wide range of tasks, then document these methods to build up a library of tools and a set of tutorials that allow others to learn prompt engineering or simply find prompts that would be ideal for them.

APPLY FOR THIS JOB

Compensation and Benefits*

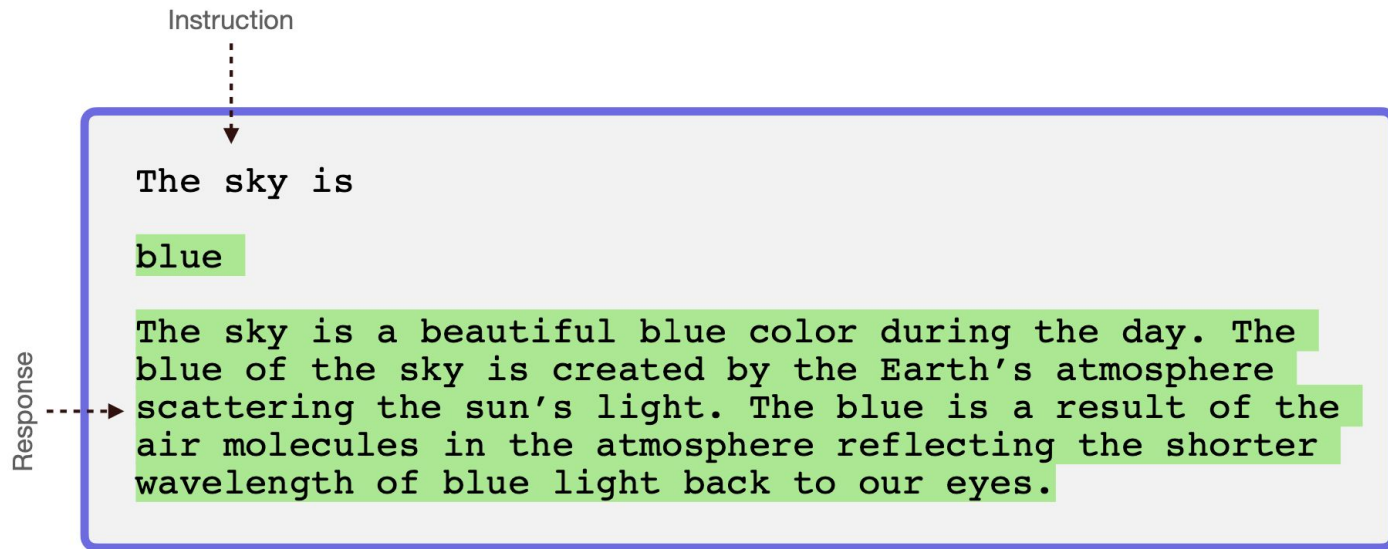
Anthropic's compensation package consists of three elements: salary, equity, and benefits. We are committed to pay fairness and aim for these three elements collectively to be highly competitive with market rates.

Salary - The expected salary range for this position is \$250k - \$335k.

Equity - Equity will be a major component of the total compensation for this position. We aim to offer higher-than-average equity compensation for a company of our size, and communicate equity amounts at the time of offer issuance.

Source: <https://jobs.lever.co/Anthropic/e3cde481-d446-460f-b576-93cab67bd1ed>

First Basic Prompt



Model: "text-davinci-003"
temperature: 0.7
top-p: 1

Elements of a Prompt

- A prompt is composed with the following components:

- Instructions
- Context
- Input data
- Output indicator

Classify the text into neutral, negative or positive

Text: I think the food was okay.

Sentiment:

Settings to keep in mind

- When prompting a language model you should keep in mind a few settings
- You can get very different results with prompts when using different settings
- One important setting is controlling how deterministic the model is when generating completion for prompts
 - **Temperature** and **top_p** are two important parameters to keep in mind
 - Generally, keep these low if you are looking for exact answers
 - ...keep them high if you are looking for more diverse responses

Controlling Model Output

- temperature and top_p control the randomness of the output.

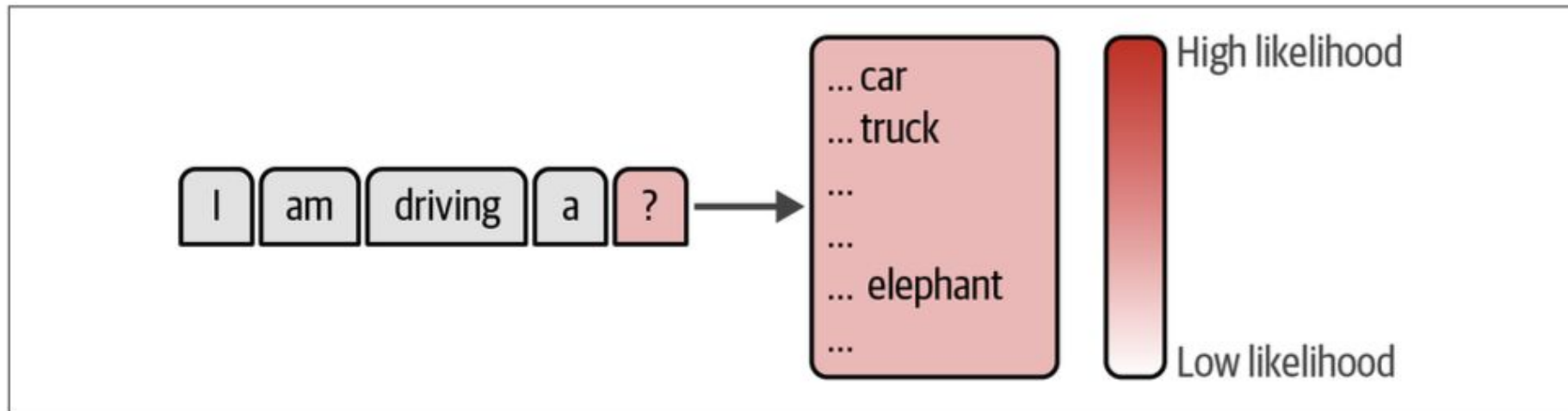


Figure 6-3. The model chooses the next token to generate based on their likelihood scores.

Temperature

- how likely it is to choose tokens that are less probable
 - 0: generates the same response every time, i.e., always choose the most likely word

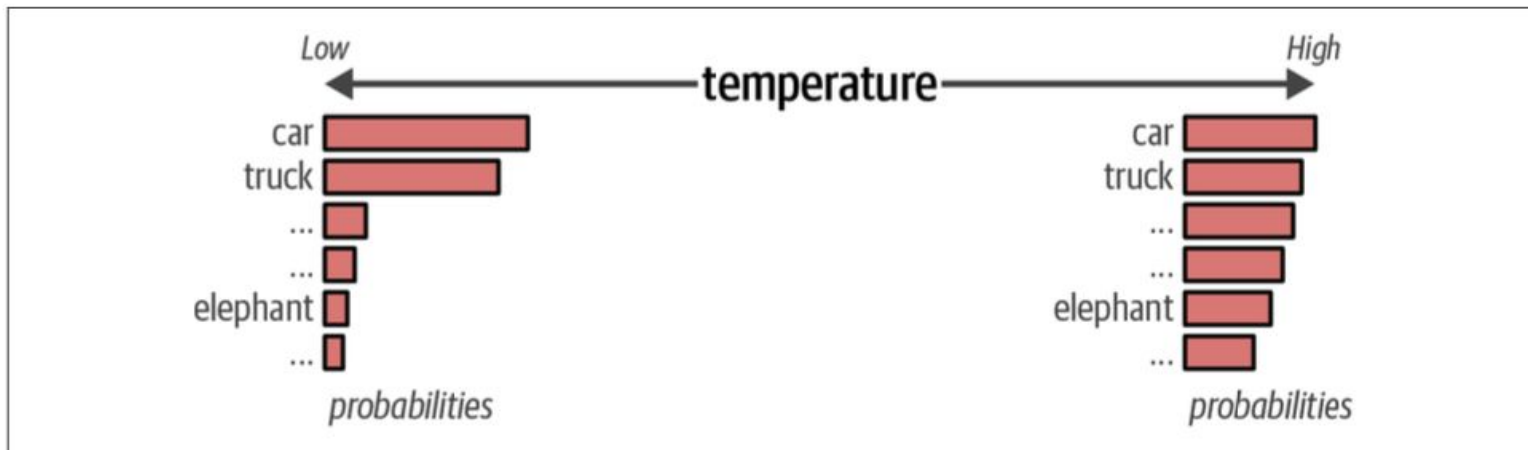


Figure 6-4. A higher temperature increases the likelihood that less probable tokens are generated and vice versa.

top_p (a.k.a nucleus sampling)

- controls which subset of tokens (the nucleus) the LLM can consider.
- E.g., $\text{top}_p = 0.1$: consider tokens until it reaches 0.1 as cumulative probability.

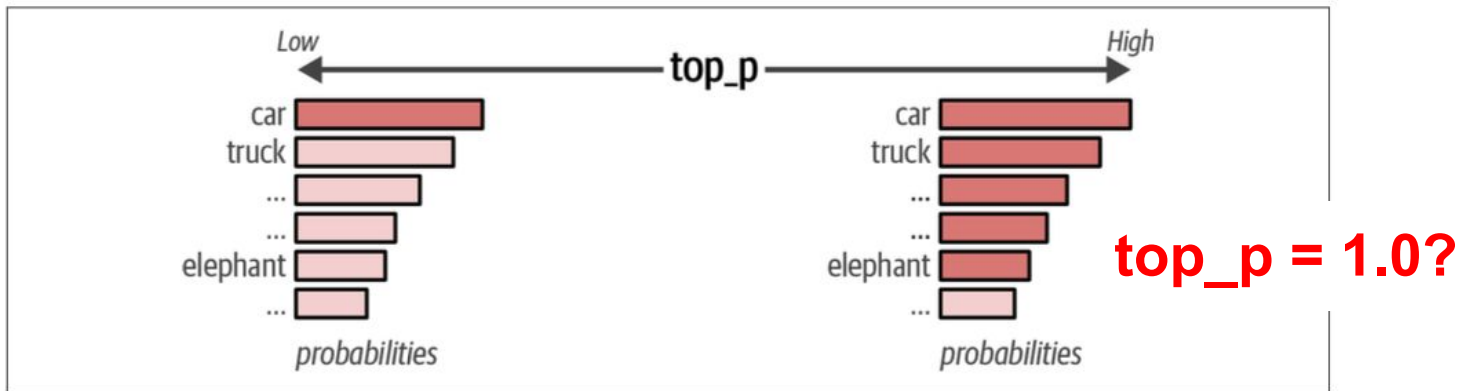


Figure 6-5. A higher top_p increases the number of tokens that can be selected to generate and vice versa.

Use cases

Table 6-1. Use case examples when selecting values for temperature and top_p.

Example use case	Temperature	top_p	Description
Brainstorming session	High	High	High randomness with large pool of potential tokens. The results will be highly diverse, often leading to very creative and unexpected results.
Email generation	Low	Low	Deterministic output with high probable predicted tokens. This results in predictable, focused, and conservative outputs.
Creative writing	High	Low	High randomness with a small pool of potential tokens. This combination produces creative outputs but still remains coherent.
Translation	Low	High	Deterministic output with high probable predicted tokens. Produces coherent output with a wider range of vocabulary, leading to outputs with linguistic variety.

Designing Prompts for Different Tasks

- In the next few slides, we will cover a few examples of common tasks using different prompts
- Instruction-based prompting:
 - Text Summarization
 - Question Answering
 - Text Classification
 - Role Playing
 - Code Generation
 - Reasoning

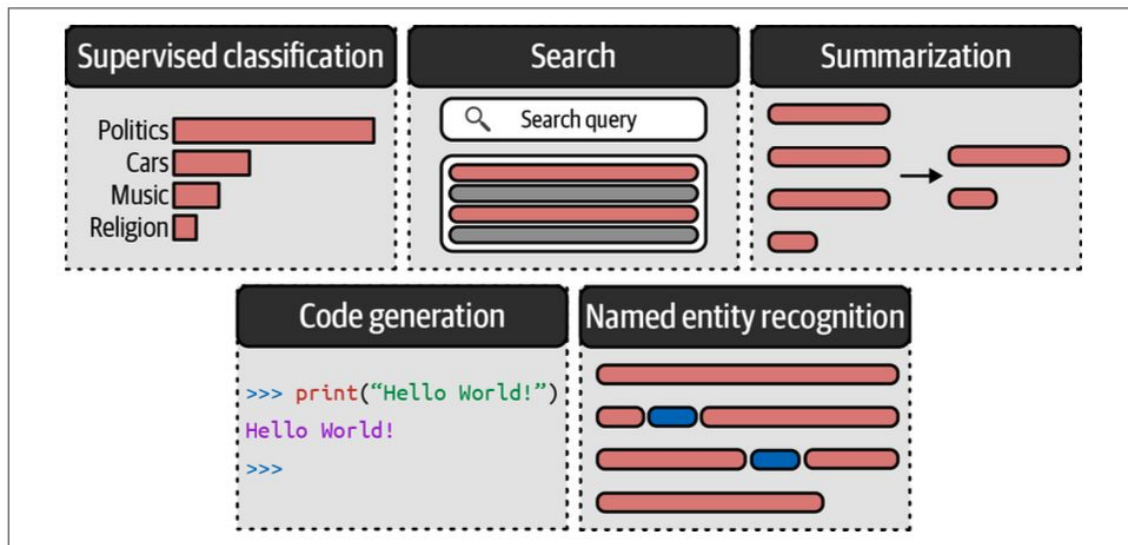


Figure 6-9. Use cases for instruction-based prompting.

Text Summarization

Context

Antibiotics are a type of medication used to treat bacterial infections. They work by either killing the bacteria or preventing them from reproducing, allowing the body's immune system to fight off the infection.

Antibiotics are usually taken orally in the form of pills, capsules, or liquid solutions, or sometimes administered intravenously. They are not effective against viral infections, and using them inappropriately can lead to antibiotic resistance.

Instruction

Explain the above in one sentence:

Antibiotics are medications used to treat bacterial infections by either killing the bacteria or stopping them from reproducing, but they are not effective against viruses and overuse can lead to antibiotic resistance.

Question Answering

Answer the question based on the context below. Keep the answer short and concise. Respond "Unsure about answer" if not sure about the answer.

Context: Teplizumab traces its roots to a New Jersey drug company called Ortho Pharmaceutical. There, scientists generated an early version of the antibody, dubbed OKT3. Originally sourced from mice, the molecule was able to bind to the surface of T cells and limit their cell-killing potential. In 1986, it was approved to help prevent organ rejection after kidney transplants, making it the first therapeutic antibody allowed for human use.

Question: What was OKT3 originally sourced from?

Answer: Mice.

Text Classification

Classify the text into neutral, negative or positive.

Text: I think the food was okay.

Sentiment: Neutral

Role Playing

The following is a conversation with an AI research assistant. The assistant tone is technical and scientific.

Human: Hello, who are you?

AI: Greeting! I am an AI research assistant. How can I help you today?

Human: Can you tell me about the creation of blackholes?

AI: Sure! Black holes are regions of spacetime where the gravitational force is so strong that nothing, not even light, can escape from it. They are created when a very massive star dies and its core collapses in on itself, forming a singularity of infinite density. The intense gravity of the singularity pulls in all the matter and radiation around it, creating the black hole.

Code Generation

```
"""
```

```
Table departments, columns = [DepartmentId,  
DepartmentName]
```

```
Table students, columns = [DepartmentId, StudentId,  
StudentName]
```

```
Create a MySQL query for all students in the Computer  
Science Department
```

```
"""
```

```
SELECT StudentId, StudentName
```

```
FROM students
```

```
WHERE DepartmentId IN (SELECT DepartmentId FROM  
departments WHERE DepartmentName = 'Computer Science');
```


Reasoning

The odd numbers in this group add up to an even number: 15, 32, 5, 13, 82, 7, 1.

Solve by breaking the problem into steps. First, identify the odd numbers, add them, and indicate whether the result is odd or even.

Odd numbers: 15, 5, 13, 7, 1

Sum: 41

41 is an odd number.

Introduction to Prompt Engineering

Demo

Part 2

Advanced Techniques for Prompt Engineering

Advanced Prompt Engineering

On the surface

- ask a specific question

- be accurate

- add some examples

- and you are done!

However, prompting can grow complex quite quickly

The potential complexity of a Prompt

Persona

Describe what role the LLM should take on. For example, use “You are an expert in astrophysics” if you want to ask a question about astrophysics.

Instruction

The task itself. Make sure this is as specific as possible. We do not want to leave much room for interpretation.

Context

Additional information describing the context of the problem or task. It answers questions like “What is the reason for the instruction?”

The potential complexity of a Prompt

Format

The format the LLM should use to output the generated text. Without it, the LLM will come up with a format itself, which is troublesome in automated systems.

Audience

The target of the generated text. This also describes the level of the generated output. For education purposes, it is often helpful to use ELI5 ("Explain it like I'm 5").

Tone

The tone of voice the LLM should use in the generated text. If you are writing a formal email to your boss, you might not want to use an informal tone of voice.

Data

The main data related to the task itself.

The summarization prompt

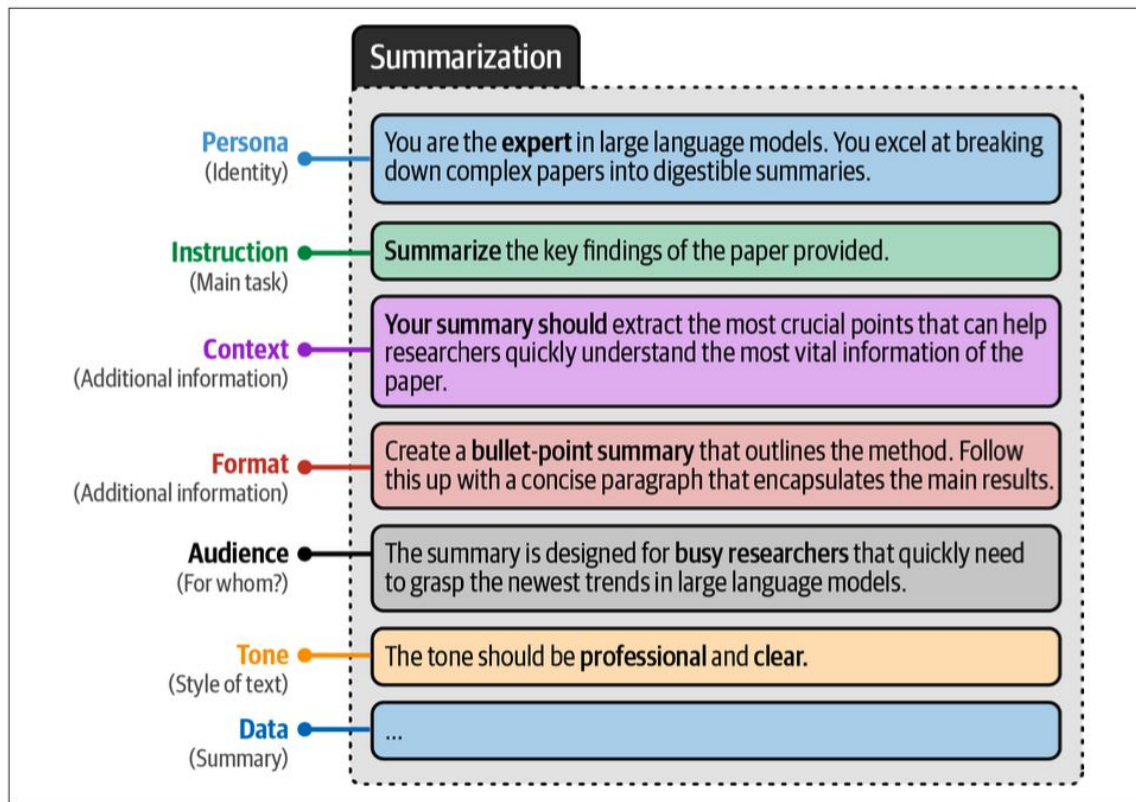


Figure 6-11. An example of a complex prompt with many components.

Introduce, remove, change order, ...

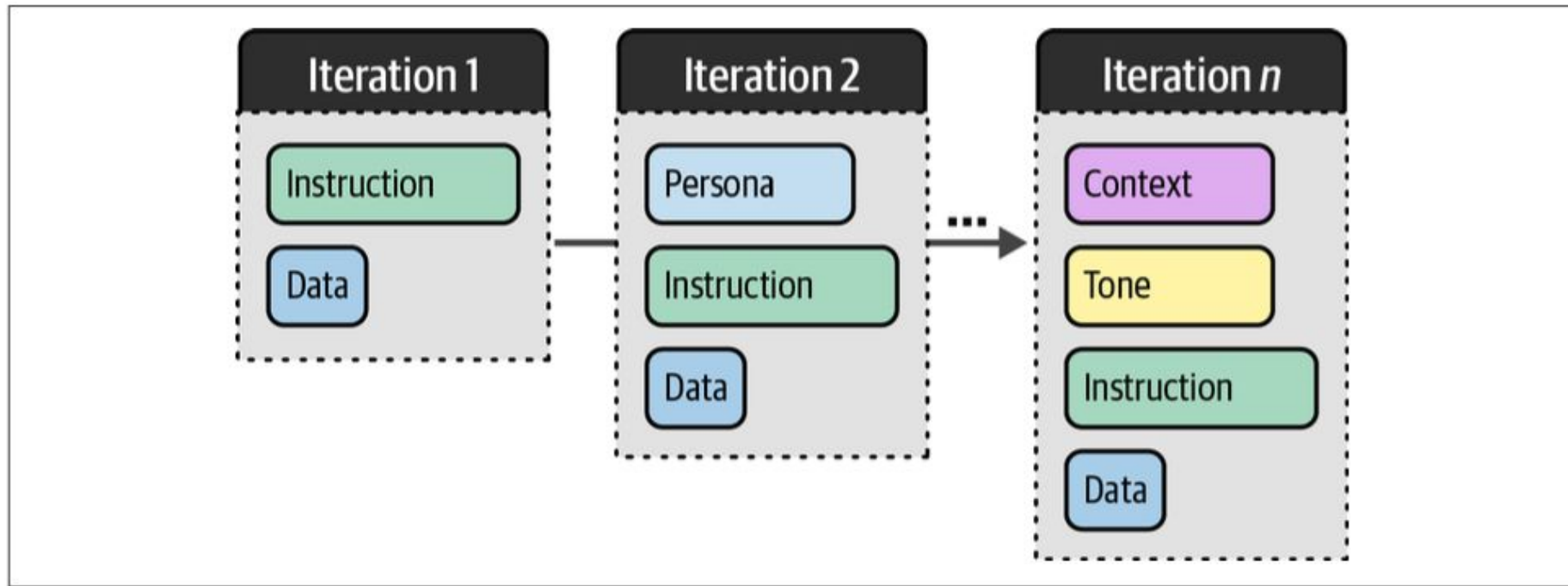


Figure 6-12. Iterating over modular components is a vital part of prompt engineering.

Providing examples

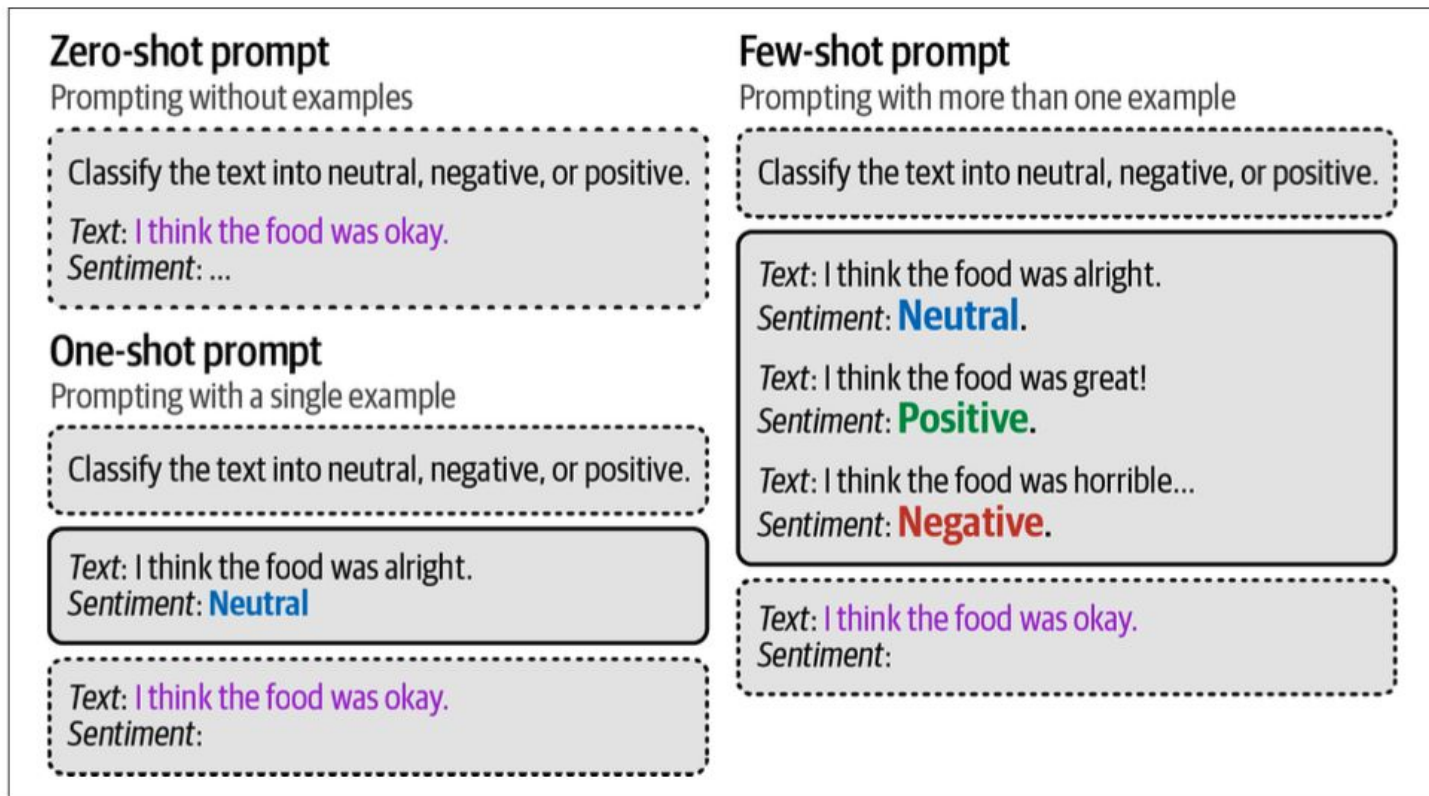


Figure 6-13. An example of a complex prompt with many components.

Chain Prompting: Breaking up the Problem

Breaking the problem **between** prompts.

Take the output of one prompt and use it as input for the next.

Creates a continuous chain of interactions to solve the problem.

Ex:

Input: product features

Output: product name, slogan, and sales pitch

Sequential pipeline

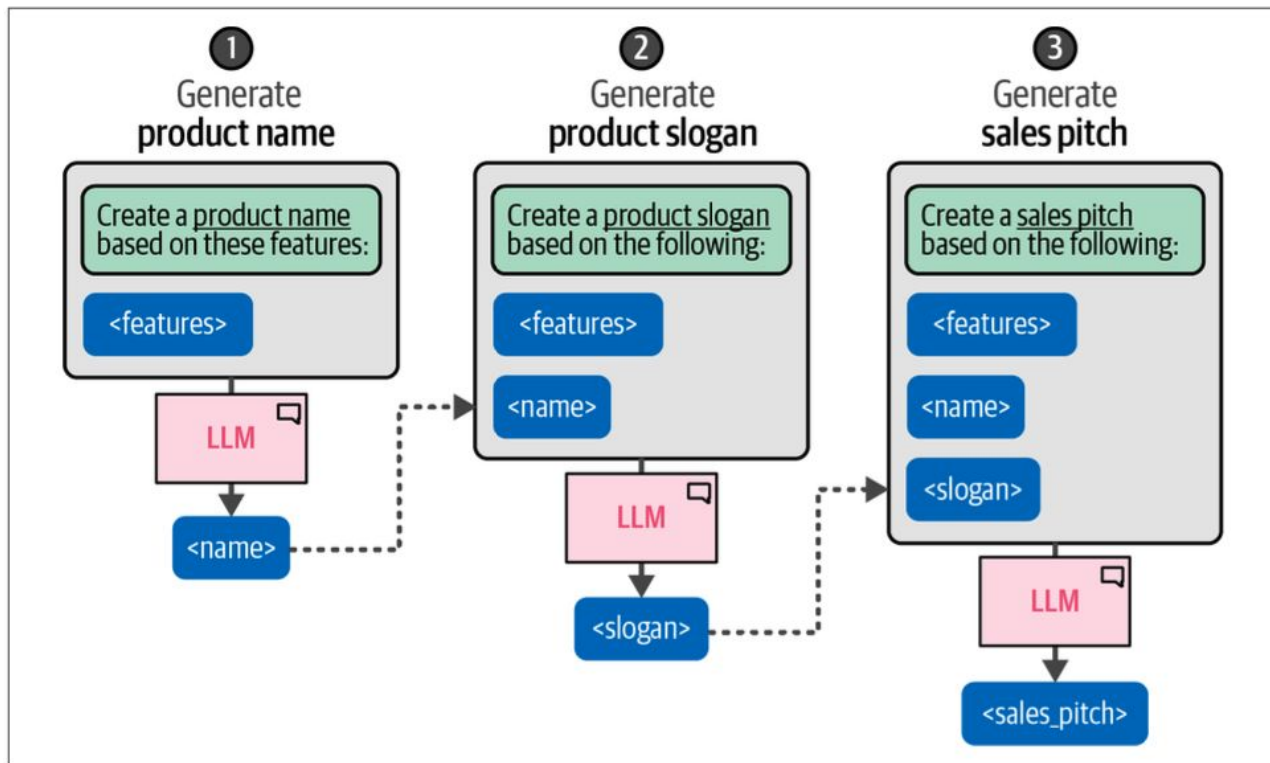
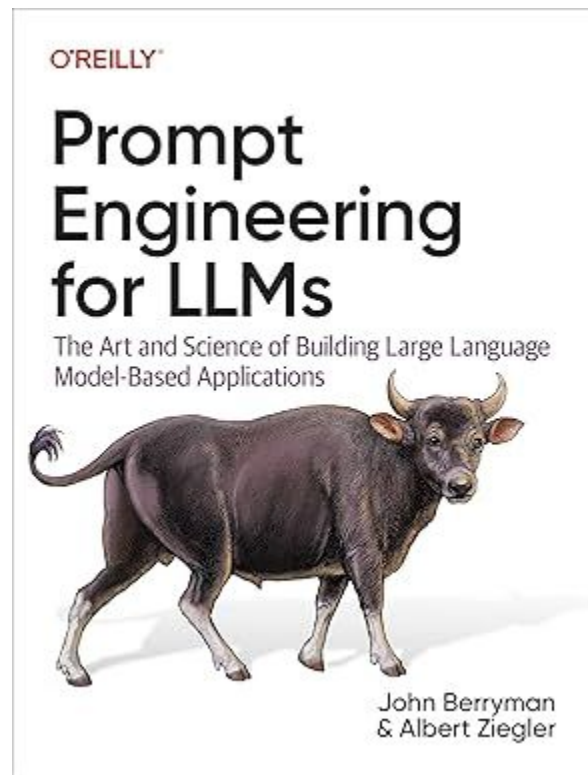
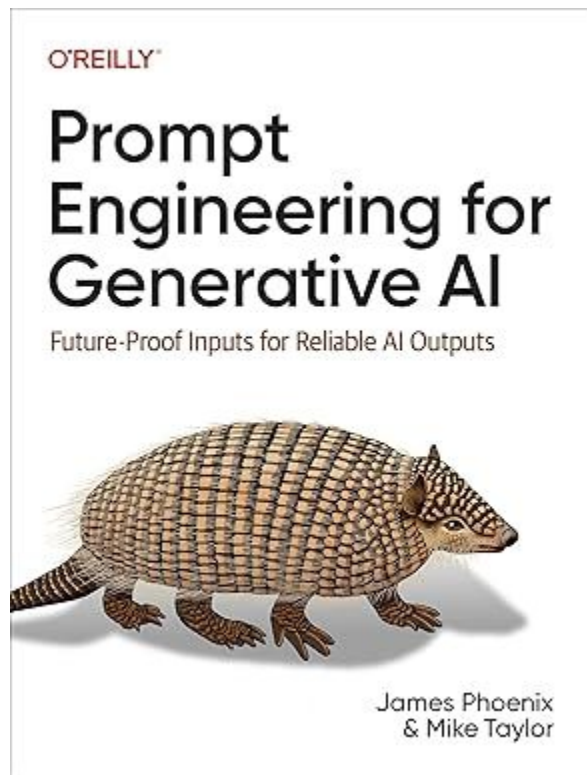


Figure 6-14. Using a description of a product's features, chain prompts to create a suitable name, slogan, and sales pitch.

Deep dive into Prompting Engineering



Reasoning with Generative Models

What do you expect from AI?

Self-driving cars

Digital assistant

Solving hardest math problems

...



A little expectation on AI

AI should be able to learn from only a few examples, like what humans do

Does machine learning meet this expectation?

Semi-supervised learning

Active learning

Transfer learning

Metalearning

Kernel machines

Manifold learning

...



What is missing in machine learning?

Reasoning

Humans can learn from a few examples because humans can reason

Humans vs Machines

Humans can handle a lot of tasks, even with only a few examples.

Few-shot Learning Ability

Humans possess the ability to generalize beyond familiar scenarios.

Out-of-distribution Robustness

Humans are capable of providing explanations for their decisions or predictions whereas machine (specifically deep neural networks) are often considered "black box" with limited explainability.

Explainability

Humans are able to **reason**, while **reasoning** is often considered **missed** in **machines**.

What's Reasoning?



What's Reasoning?



Reasoning refers to the process of thinking through a problem or situation in order to form a logical conclusion or make a decision. It involves using evidence, facts, and logic to arrive at a solution or answer. Reasoning can be deductive, inductive or abductive and can be performed using formal or informal methods.

Reasoning refers to the process of thinking through a problem or situation in order to form a logical conclusion or make a decision. It involves using evidence, facts, and logic to arrive at a solution or answer. Reasoning can be **deductive, inductive or abductive** and can be performed using **formal** or **informal** methods.

Common Types of Reasoning

Deductive Reasoning:

Premise: All mammals have kidneys.

Premise: All whales are mammals.

Conclusion: All whales have kidneys.

Inductive Reasoning:

Observation: Every time we see a creature with wings, it is a bird.

Observation: We see a creature with wings.

Conclusion: The creature is likely to be a bird.

Abductive Reasoning, analogical reasoning, causal reasoning, probabilistic reasoning...

Formal Reasoning vs ***Informal*** Reasoning

Formal Reasoning is a systematic and logical process that follows a set of rules and principles, often used in mathematics and logic.

more structured and reliable

Informal Reasoning is a less structured approach that relies on intuition experience, and common sense to draw conclusions and solve problems, and is often used in everyday life.

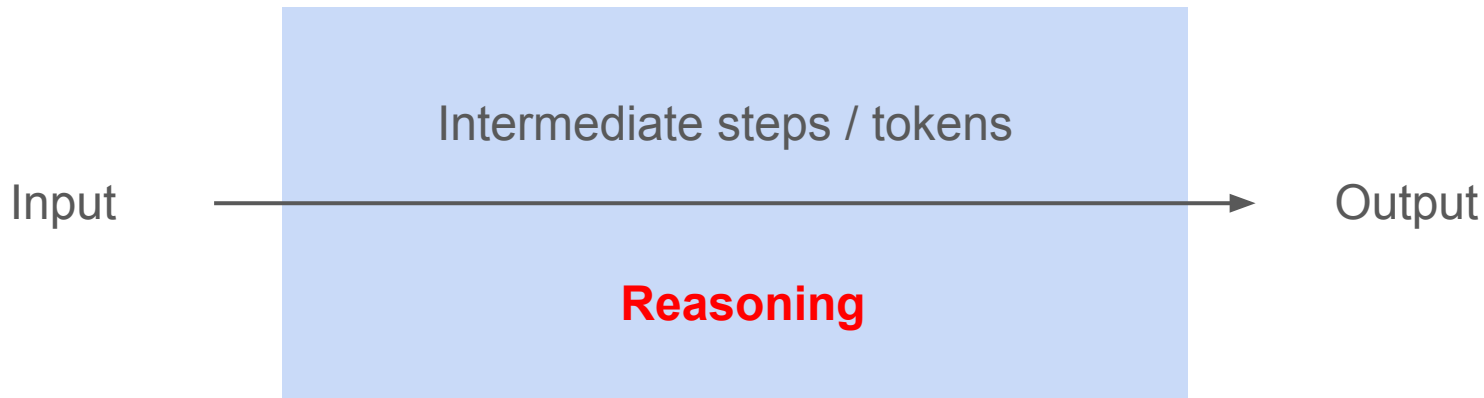
more adaptable and open-ended

Reasoning in Language Models

usually no clear definition

focus on **informal deductive reasoning** – a widely used form in which the conclusion is guaranteed to be true as long as the premises are true

What is LLM Reasoning?



Ling et al. Program Induction by Rationale Generation: Learning to Solve and Explain Algebraic Word Problems. ACL. 2017.
Chen et al. Compositional Generalization via Neural-Symbolic Stack Machines. NeurIPS. 202.

We have found a simple way to solve reasoning

**Teach language models to reason,
like teaching kids**

Rationale generation

Introduced natural language to solve math problems.

Given (i) the problem and (ii) the set of options

Predict (i) the rationale and (ii) the correct option

Problem 1:

Question: Two trains running in opposite directions cross a man standing on the platform in 27 seconds and 17 seconds respectively and they cross each other in 23 seconds. The ratio of their speeds is:

Options: A) $3/7$ B) $3/2$ C) $3/88$ D) $3/8$ E) $2/2$

Rationale: Let the speeds of the two trains be x m/sec and y m/sec respectively. Then, length of the first train = $27x$ meters, and length of the second train = $17y$ meters. $(27x + 17y) / (x + y) = 23 \rightarrow 27x + 17y = 23x + 23y \rightarrow 4x = 6y \rightarrow x/y = 3/2$.

Correct Option: B

Input

Output

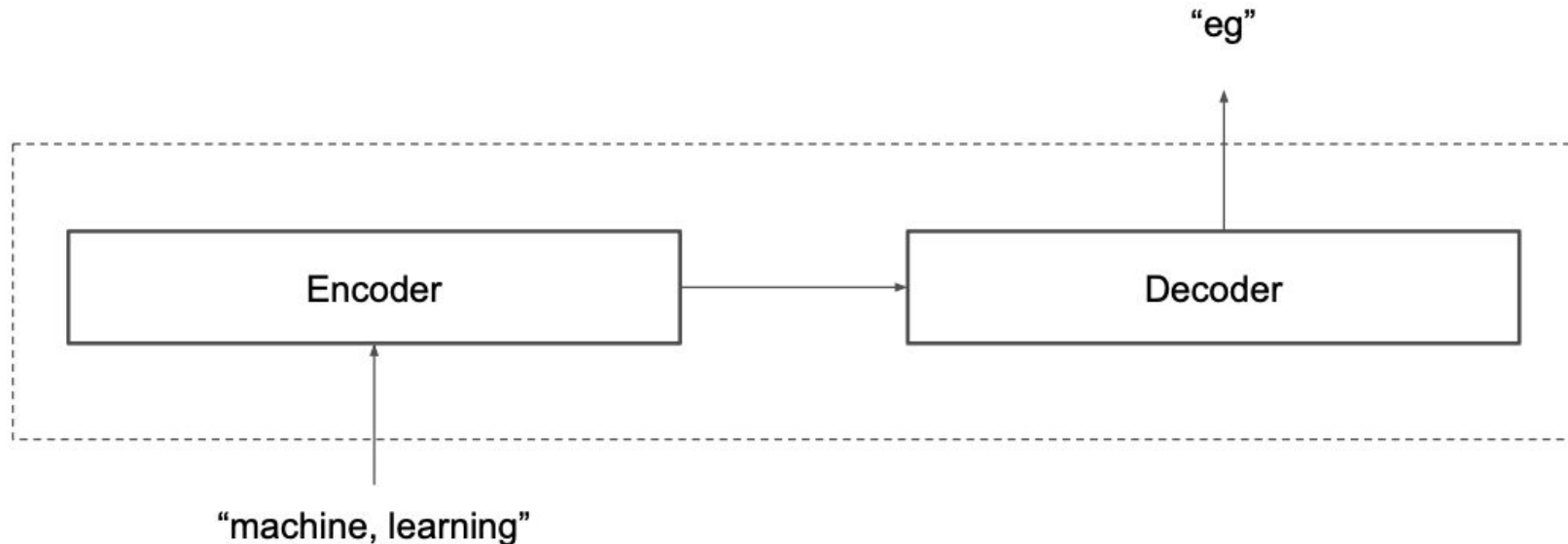
Let's start from a toy problem

A toy machine learning problem: last-letter-concatenation

Input	Output
"Elon Musk"	"nk"
"Bill Gates"	"ls"

Rule: Take the last letter of each word, and then concatenate them.

Solve it by machine learning? Tons of labels needed

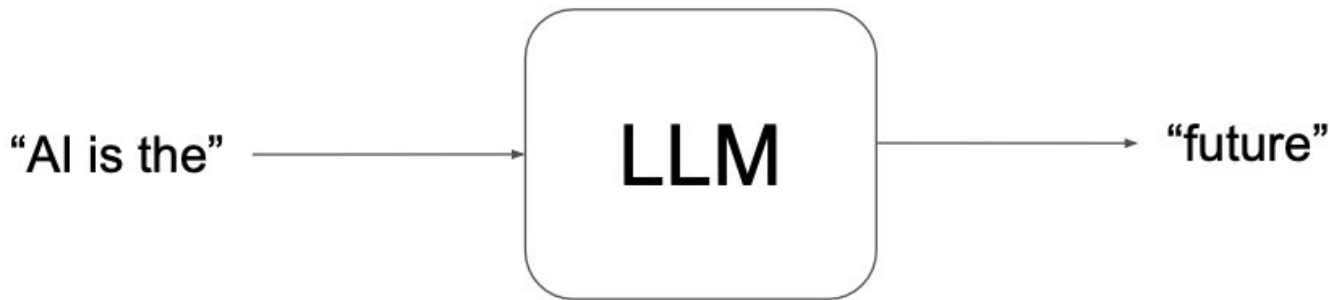


Would you like to call an ML method which needs tons of labels to learn a *trivial* task as AI?

How to solve this problem with LLMS?

What are Large Language Models (LLMs)?

LLM is a "transformer" model trained to predict the next word

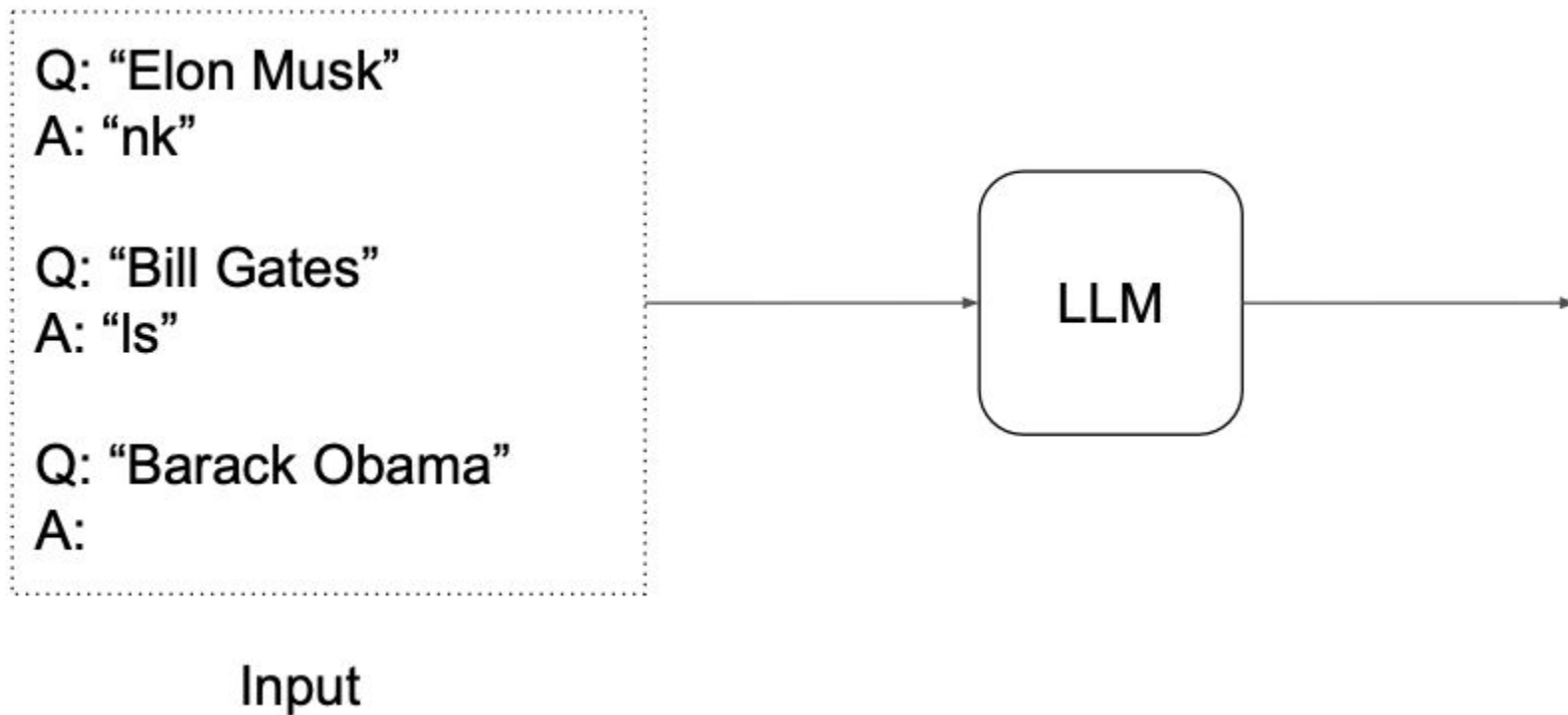


Trained on many sentences, e.g. all texts from the Internet

**One can think of training
an LLM as training a
parrot to mimic human
languages**



Few-shot prompting for last-letter-concatenation



Q: "Elon Musk"

A: "nk"

Q: "Bill Gates"

A: "Is"

Q: "Barack Obama"

A: "ma"

FAILED

Complete

Model

text-davinci-003

Temperature 0

Maximum length 256

Stop sequences

Enter sequence and press Tab

Top P 1

Frequency penalty 0

Presence penalty 0

Submit



54

Q: "Elon Musk"

A: "nk"

Q: "Bill Gates"

A: "Is"

Q: "Barack Obama"

A: "ma"



☰ Complete ▾

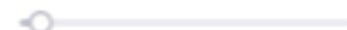
Model

text-davinci-003 ▾

Temperature 0



Maximum length 256



Stop sequences

Enter sequence and press Tab

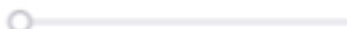
Top P 1



Frequency penalty 0



Presence penalty 0



How about adding more examples?

Submit



54

Q: "Elon Musk"

A: "nk"

Q: "Bill Gates"

A: "ls"

Q: "Steve Jobs"

A: "es"

Q: "Larry Page"

A: "ye"

Q: "Jeff Bezos"

A: "fs"

Q: "Barack Obama"

A: "ma"

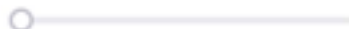
FAILED

☰ Complete ▾

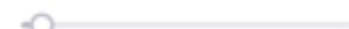
Model

text-davinci-003 ▾

Temperature 0



Maximum length 256



Stop sequences

Enter sequence and press Tab

Top P 1



Frequency penalty 0



Presence penalty 0



Submit



94

Let's add "reasoning process" before "answer"

Q: "Elon Musk"

A: the last letter of "Elon" is "n". the last letter of "Musk" is "k". Concatenating "n", "k" leads to "nk". so the output is "nk".

reasoning process

Q: "Bill Gates"

A: the last letter of "Bill" is "l". the last letter of "Gates" is "s". Concatenating "l", "s" leads to "ls". so the output is "ls".

Q: "Barack Obama"

A:

Let's add "reasoning process" before "answer"

Q: "Elon Musk"

A: the last letter of "Elon" is "n". the last letter of "Musk" is "k". Concatenating "n", "k" leads to "nk". so the output is "nk".

reasoning process

Q: "Bill Gates"

A: the last letter of "Bill" is "l". the last letter of "Gates" is "s". Concatenating "l", "s" leads to "ls". so the output is "ls".

Q: "Barack Obama"

A: the last letter of "Barack" is "k". the last letter of "Obama" is "a". Concatenating "k", "a" leads to "ka". so the output is "ka".

One demonstration is enough, like humans

Q: "Elon Musk"

A: the last letter of "Elon" is "n". the last letter of "Musk" is "k". Concatenating "n", "k" leads to "nk". so the output is "nk".

Q: "Barack Obama"

A: the last letter of "Barack" is "k". the last letter of "Obama" is "a". Concatenating "k", "a" leads to "ka". so the output is "ka".

100% accuracy with only one demonstration example

Why the last-letter-concatenation task?

Machine learning fails

Few-shot prompting fails

Trivial for humans



Studio



Chat

Stream

Generate media

Build

History



Dashboard



Documentation

User

Q: Elon Musk

A: nk

Q: Bill Gates

A: Is

Q: Barack Obama

A:

Model

✦ Thoughts (experimental)

Manual: 32768

Expand to view model thoughts



ka

Google Search Suggestions

Display of Search Suggestions is required when using Grounding with Google Search. [Learn more](#)

Barack Obama

pattern nk Is Elon Musk Bill Gates

Google AI models may make mistakes, so double-check outputs.

Get API key

Settings

anisiom@gmail.com

Gemini 2.5 Pro

gemini-2.5-pro

Our most powerful reasoning model, which excels at coding and complex reasoning tasks.

System instructions

Optional tone and style instructions for the model

Temperature



1

Media resolution

Default

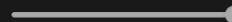


Thinking

Thinking mode



Set thinking budget



3276

Tools



Structured output

Edit



Code execution



Function calling

Edit



Start typing a prompt



Run

Analyzing User Requests

I've initiated the first step, aiming to understand the user's request and the structural patterns within the example inputs. My current focus is on a Google search to decode the prompt.

Discovering the Code

I'm now conducting a Google search for "Barack Obama," hoping to find information that fits the initial pattern I've observed. The goal is to identify the relevant two-letter code associated with this individual. After that, I will analyze the search results to ascertain the applicable code.

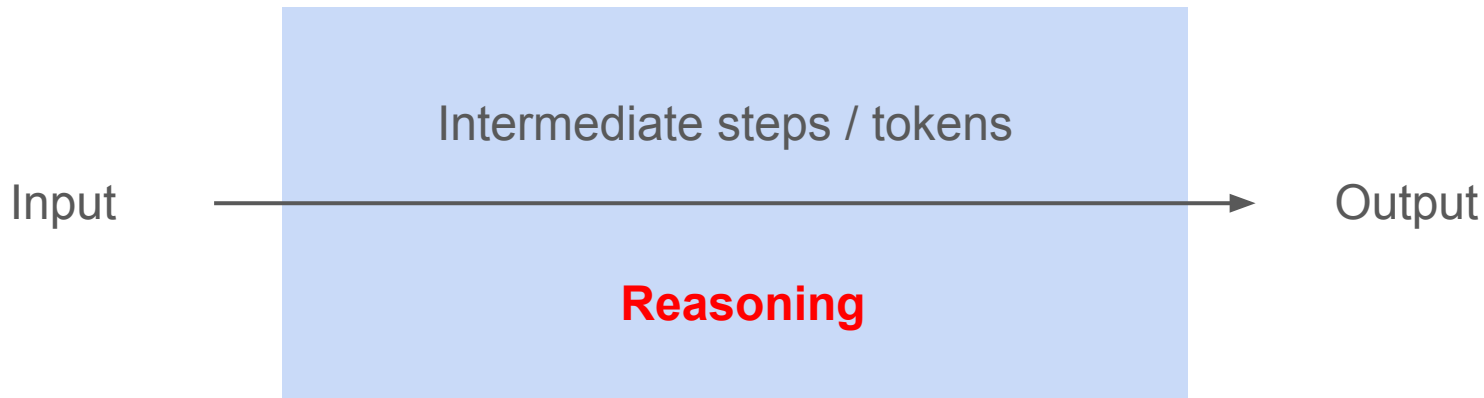
Deduce the Pattern

I've been examining the user's examples, specifically the provided inputs and outputs. I'm trying to decipher the underlying rule that transforms the names into those seemingly random sets of letters. So far, no obvious pattern is emerging. I need to dig deeper. I'm focusing on trying to identify characteristics, rules or algorithms that might be in play here. I'm trying to see if there is a pattern I'm missing!

Pinpoint the Algorithm

I think I've cracked the code! By focusing on the examples, I realized the answer boils down to the last letters of each name component. "Elon Musk" maps to "nk," and "Bill Gates" to "Is." It seems straightforward - apply this to "Barack Obama," and we should get "ka." I'm confident in this, no further digging necessary.

What is LLM Reasoning?



Ling et al. Program Induction by Rationale Generation: Learning to Solve and Explain Algebraic Word Problems. ACL. 2017.
Chen et al. Compositional Generalization via Neural-Symbolic Stack Machines. NeurIPS. 202.

Key Idea: Derive the Final Answer through **Intermediate Steps**

Ling et al 2017 pioneered using **natural language rationale** to solve math problems by "... *derive the final answer through a series of small steps*". Trained a seq-2-seq model from scratch.

Problem 1:

Question: Two trains running in opposite directions cross a man standing on the platform in 27 seconds and 17 seconds respectively and they cross each other in 23 seconds. The ratio of their speeds is:

Options: A) $3/7$ B) $3/2$ C) $3/88$ D) $3/8$ E) $2/2$

Rationale: Let the speeds of the two trains be x m/sec and y m/sec respectively. Then, length of the first train = $27x$ meters, and length of the second train = $17y$ meters. $(27x + 17y) / (x + y) = 23 \rightarrow 27x + 17y = 23x + 23y \rightarrow 4x = 6y \rightarrow x/y = 3/2$.

Correct Option: B

		Question	Rationale
Training Examples		100,949	
Dev Examples		250	
Test Examples		250	
Numeric	Average Length	9.6	16.6
	Vocab Size	21,009	14,745
Non-Numeric	Average Length	67.8	89.1
	Vocab Size	17,849	25,034
All	Average Length	77.4	105.7
	Vocab Size	38,858	39,779

Table 1: Descriptive statistics of our dataset.

GSM8K: <Problem, Intermediate Steps, Answer>

Following the work by Ling et al 2017, Cobbe et al 2021 in OpenAI built a much larger math word problem dataset (**GSM8K**) with natural language rationales, and used it to **finetune GPT3**.

GSM

Fol

muc

lang

a

Problem: Beth bakes 4, 2 dozen batches of cookies in a week. If these cookies are shared amongst 16 people equally, how many cookies does each person consume?

Solution: Beth bakes 4 2 dozen batches of cookies for a total of $4 \times 2 = 8$ dozen cookies

There are 12 cookies in a dozen and she makes 8 dozen cookies for a total of $12 \times 8 = 96$ cookies

She splits the 96 cookies equally amongst 16 people so they each eat $96/16 = 6$ cookies

Final Answer: 6

Problem: Mrs. Lim milks her cows twice a day. Yesterday morning, she got 68 gallons of milk and in the evening, she got 82 gallons. This morning, she got 18 gallons fewer than she had yesterday morning. After selling some gallons of milk in the afternoon, Mrs. Lim has only 24 gallons left. How much was her revenue for the milk if each gallon costs \$3.50?

Mrs. Lim got 68 gallons - 18 gallons = 50 gallons this morning.

So she was able to get a total of 68 gallons + 82 gallons + 50 gallons = 200 gallons.

She was able to sell 200 gallons - 24 gallons = 176 gallons.

Thus, her total revenue for the milk is $\$3.50/\text{gallon} \times 176 \text{ gallons} = \616 .

Final Answer: 616

Problem: Tina buys 3 12-packs of soda for a party. Including Tina, 6 people are at the party. Half of the people at the party have 3 sodas each, 2 of the people have 4, and 1 person has 5. How many sodas are left over when the party is over?

Solution: Tina buys 3 12-packs of soda, for $3 \times 12 = 36$ sodas

6 people attend the party, so half of them is $6/2 = 3$ people

Each of those people drinks 3 sodas, so they drink $3 \times 3 = 9$ sodas

Two people drink 4 sodas, which means they drink $2 \times 4 = 8$ sodas

With one person drinking 5, that brings the total drank to $5 + 9 + 8 + 3 = 25$ sodas

As Tina started off with 36 sodas, that means there are $36 - 25 = 11$ sodas left

Final Answer: 11

Figure 1: Three example problems from GSM8K. Calculation annotations are highlighted in red.

Show Your Work: Scratchpads for Intermediate Computation with Language Models

```
Input:
2 9 + 5 7

Target:
<scratch>
2 9 + 5 7 , C: 0
2 + 5 , 6 C: 1 # added 9 + 7 = 6 carry 1
, 8 6 C: 0 # added 2 + 5 + 1 = 8 carry 0
0 8 6
</scratch>
8 6
```

Figure 2: Example of input and target for addition with a scratchpad. The carry is recorded in the digit following “C:”. Comments (marked by #) are added for clarity and are not part of the target.

Chain-of-Thought (CoT) Prompting

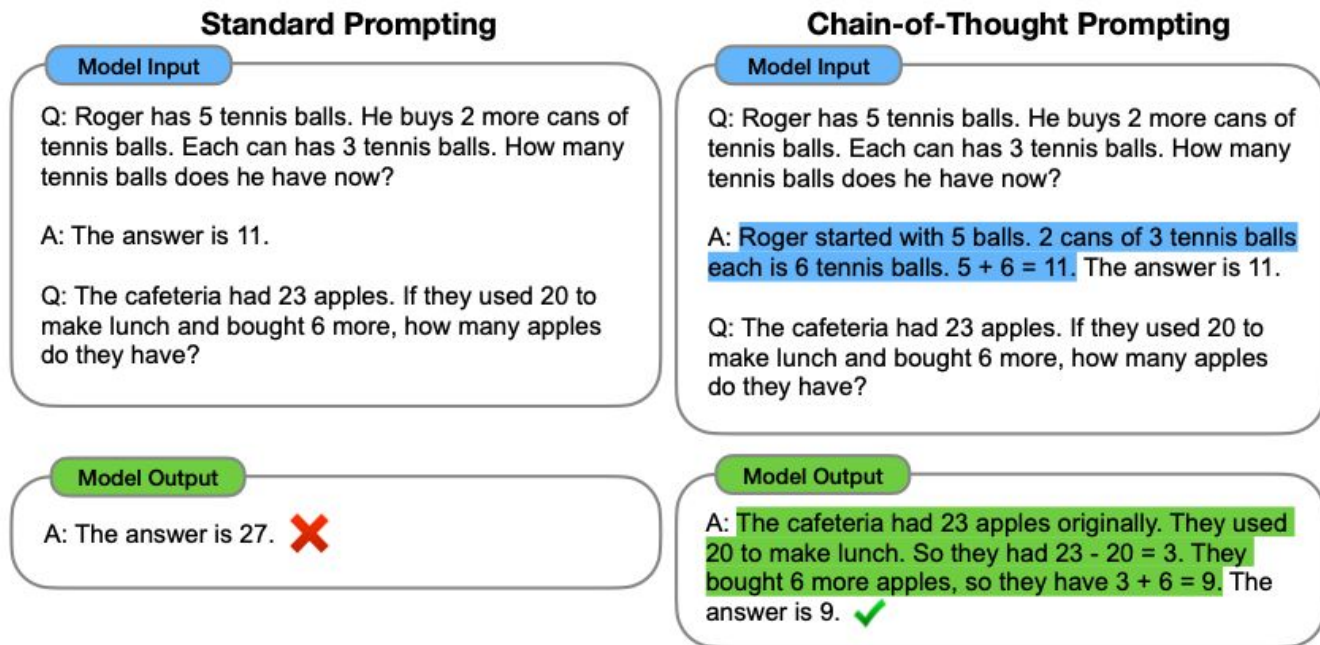


Figure 1: Chain-of-thought prompting enables large language models to tackle complex arithmetic, commonsense, and symbolic reasoning tasks. Chain-of-thought reasoning processes are highlighted.

Chain-of-Thought (CoT) Prompting

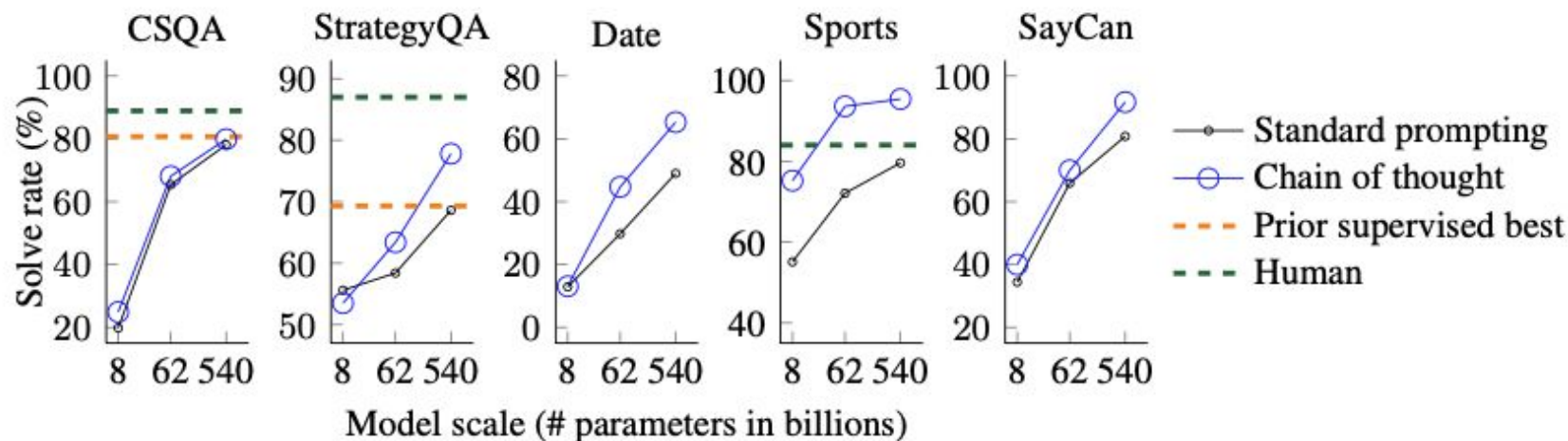


Figure 7: Chain-of-thought prompting also improves the commonsense reasoning abilities of language models. The language model shown here is PaLM. Prior best numbers are from the leaderboards of CSQA (Talmor et al., 2019) and StrategyQA (Geva et al., 2021) (single-model only, as of May 5, 2022). Additional results using various sizes of LaMDA, GPT-3, and PaLM are shown in Table 4.

Training with intermediate steps (Ling et al 2017)

Finetuning with intermediate steps (Cobbe et al 2021)

Prompting with intermediate steps (Nye et al 2021, Wei et al 2022)

~~Training~~ with intermediate steps (Ling et al 2017)

~~Finetuning~~ with intermediate steps (Cobbe et al 2021)

~~Prompting~~ with intermediate steps (Nye et al 2021, Wei et al 2022)

This is what really matters!

Regardless of training, fine-tuning, or prompting, when provided with examples that include intermediate steps, LLMs will respond with intermediate steps

**Is it also helpful to add reasoning
strategies in demonstration
examples?**

Stage 1: Decompose Question into Subquestions

Q: It takes Amy 4 minutes to climb to the top of a slide. It takes her 1 minute to slide down. The water slide closes in 15 minutes. How many times can she slide before it closes?

Language Model

A: To solve "How many times can she slide before it closes?", we need to first solve: "How long does each trip take?"

Stage 2: Sequentially Solve Subquestions

Subquestion 1

It takes Amy 4 minutes to climb to the top of a slide. It takes her 1 minute to slide down. The slide closes in 15 minutes.

Q: How long does each trip take?

Language Model

A: It takes Amy 4 minutes to climb and 1 minute to slide down. $4 + 1 = 5$. So each trip takes 5 minutes.

Append model answer to Subquestion 1

It takes Amy 4 minutes to climb to the top of a slide. It takes her 1 minute to slide down. The slide closes in 15 minutes.

Q: How long does each trip take?

A: It takes Amy 4 minutes to climb and 1 minute to slide down. $4 + 1 = 5$. So each trip takes 5 minutes.

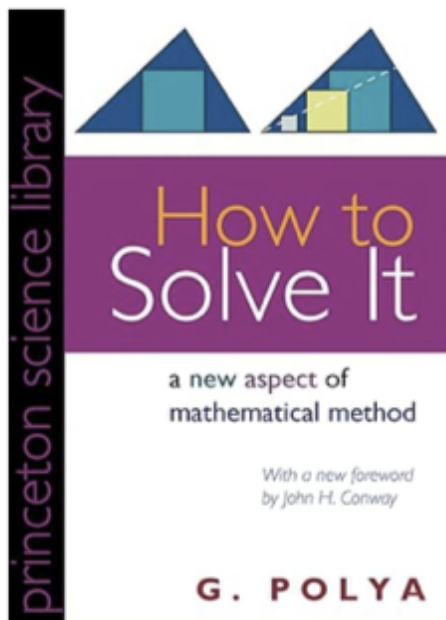
Subquestion 2

Q: How many times can she slide before it closes?

Language Model

A: The water slide closes in 15 minutes. Each trip takes 5 minutes. So Amy can slide $15 \div 5 = 3$ times before it closes.

Figure 1: Least-to-most prompting solving a math word problem in two stages: (1) query the language model to decompose the problem into subproblems; (2) query the language model to sequentially solve the subproblems. The answer to the second subproblem is built on the answer to the first subproblem. The demonstration examples for each stage's prompt are omitted in this illustration.



Decomposing and recombining are important operations of the mind.

differently. You decompose the whole into its parts, and you recombine the parts into a more or less different whole.

1. If you go into detail you may lose yourself in details. Too many or too minute particulars are a burden on the mind. They may prevent you from giving sufficient attention to the main point, or even from seeing the main point at all. Think of the man who cannot see the forest for the trees.

Q: Elsa has 5 apples. Anna has 2 more apples than Elsa. How many apples do they have together?
A: Let's break down this problem: 1. How many apples does Anna have? 2. How many apples do they have together?

1. Anna has 2 more apples than Elsa. So Anna has $2 + 5 = 7$ apples.
2. Elsa and Anna have $5 + 7 = 12$ apples together.

The answer is: 12.

Table 9: The least-to-most prompt for solving GSM8K. The demonstration problem is solved in only two steps, but the prompt can handle problems that require multiple steps to be solved.

Q: Elsa has 5 apples. Anna has 2 more apples than Elsa. How many apples do they have together?
A: Anna has 2 more apples than Elsa. So Anna has $2 + 5 = 7$ apples. So Elsa and Anna have $5 + 7 = 12$ apples together.

The answer is: 12.

Table 10: The chain-of-thought prompt for solving GSM8K. It is derived from the least-to-most prompt in Table 9 by removing the decomposition part.

Accuracy by Steps (GSM8K)	All	2 Steps	3 Steps	4 steps	≥ 5 steps
Least-to-Most	62.39	74.53	68.91	59.73	45.23
Chain-of-Thought	60.87	76.68	67.29	59.39	39.07

Table 12: Accuracies (%) of least-to-most prompting and chain-of-thought prompting, broken down by the number of reasoning steps required in the expected solution.

CFQ (Compositional Generalization): Text-to-Code

	MCD1	MCD2	MCD3	Ave.
Fully Supervised				
T5-base (Herzig et al., 2021)	58.5	27.0	18.4	34.6
T5-large (Herzig et al., 2021)	65.1	32.3	25.4	40.9
T5-3B (Herzig et al., 2021)	65.0	41.0	42.6	49.5
HPD (Guo et al., 2020)	79.6	59.6	67.8	69.0
T5-base + IR (Herzig et al., 2021)	85.8	64.0	53.6	67.8
T5-large + IR (Herzig et al., 2021)	88.6	79.2	72.7	80.2
T5-3B + IR (Herzig et al., 2021)	88.4	85.3	77.9	83.9
LeAR (Liu et al., 2021)	91.7	89.2	91.7	90.9
Prompting				
(Ours) Dynamic Least-to-Most	94.3	95.3	95.5	95.0

Using just 1% data!

Why intermediate steps are helpful?

Why "Intermediate Tokens" / "Reasoning" Matters?

LLMs are probabilistic models $\neq$ Humans

Increasing the length of CoT can drastically make transformers more expressive.

- **Constant-depth transformers** can solve any inherently serial problem as long as it generates sufficiently long intermediate reasoning steps
- **Transformers which directly generate final answers** either requires a huge depth to solve or cannot solve at all

Transformers can only express the groundtruth function by using CoT.

Practical implications of this theory

Generating more intermediate steps (think longer)

Too long to generate? Calling external tools

...

Is it possible to trigger step by step reasoning without using demonstration examples?

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: The answer (arabic numerals) is

(Output) 8 ✗

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: **Let's think step by step.**

(Output) *There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓*

Let's think step by step

Kojima, T., Gu, S.S., Reid, M., Matsuo, Y. and Iwasawa, Y. Large language models are zero-shot reasoners. NeurIPS 2022.

Hmm, zero-shot is cool but usually significantly worse than few-shot...

LLMs as Analogical Reasoners



How to Solve It

a new aspect of
mathematical method

With a new foreword
by John H. Conway

G. POLYA

Do you know a related problem? We can scarcely imagine a problem absolutely new, unlike and unrelated to any formerly solved problem; but, if such a problem could exist, it would be insoluble. In fact, when solving a problem, we always profit from previously solved problems, using their result, or their method, or the experience we acquired solving them. And, of course, the problems from which we profit must be in some way related to our present problem. Hence the question: *Do you know a related problem?*

There is usually no difficulty at all in recalling formerly solved problems which are more or less related

Analological Prompting

"Prompts language models to self-generate relevant exemplars or knowledge in the context, before proceeding to solve the given problem"

What is the area of the square with the four vertices at $(-2, 2)$, $(2, -2)$, $(-2, -6)$, and $(-6, -2)$? Recall a related problem, and then solve this one.

The related problem is finding the distance between two points on a coordinate plane. To find the distance between two points, we use the distance formula: $d = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$.

In this problem, we have four points: $(-2, 2)$, $(2, -2)$, $(-2, -6)$, and $(-6, -2)$. Let's choose $(-2, 2)$ and $(2, -2)$ to find the distance d between them: $d = \sqrt{((2 - (-2))^2 + (-2 - 2)^2)} = \sqrt{(4^2 + (-4)^2)} = \sqrt{(16 + 16)} = \sqrt{32}$. We know that the distance between two opposite sides of a square is equal. So, the length of each side of the square is $\sqrt{32}$. To find the area of the square, we square the length of one side: $\text{Area} = (\sqrt{32})^2 = 32$.

Analogical reasoner vs zero-shot / few-shot CoT

Prompting Method	GSM8K Accuracy			MATH Accuracy	
	GPT3.5-turbo	text-davinci-003	PaLM2	GPT3.5-turbo	PaLM2
0-shot	75.0%	14.8%	60.8%	33.0%	27.1%
0-shot CoT	75.8%	50.3%	78.2%	33.9%	29.8%
5-shot CoT	76.7%	54.0%	80.7%	34.9%	34.3%
Ours: Self-generated Exemplars	77.8%	61.0%[†]	81.7%	37.3%	34.8%

Prompting Method	GPT3.5-turbo-16k		GPT4	
	Acc@1	Acc@10	Acc@1	Acc@10
0-shot	8%	24%	16%	30%
0-shot CoT	9%	27%	16%	29%
3-shot CoT	11%	27%	17%	31%
Ours: Self-generated Exemplars	13%	25%	17%	32%
Ours: Self-generated Knowledge + Exemplars	15%	29%	19%	37%

Analogical reasoning: Key idea

Adaptively generate relevant examples and knowledge, rather than just using a fix set of examples.

Is it possible to trigger step by step reasoning even without using any prompt like "let's think step by step"?

Chain-of-Thought Reasoning without Prompting

Pretrained LLMs are ready to reason

All we need is decoding

Chain-of-Thought Reasoning without Prompting

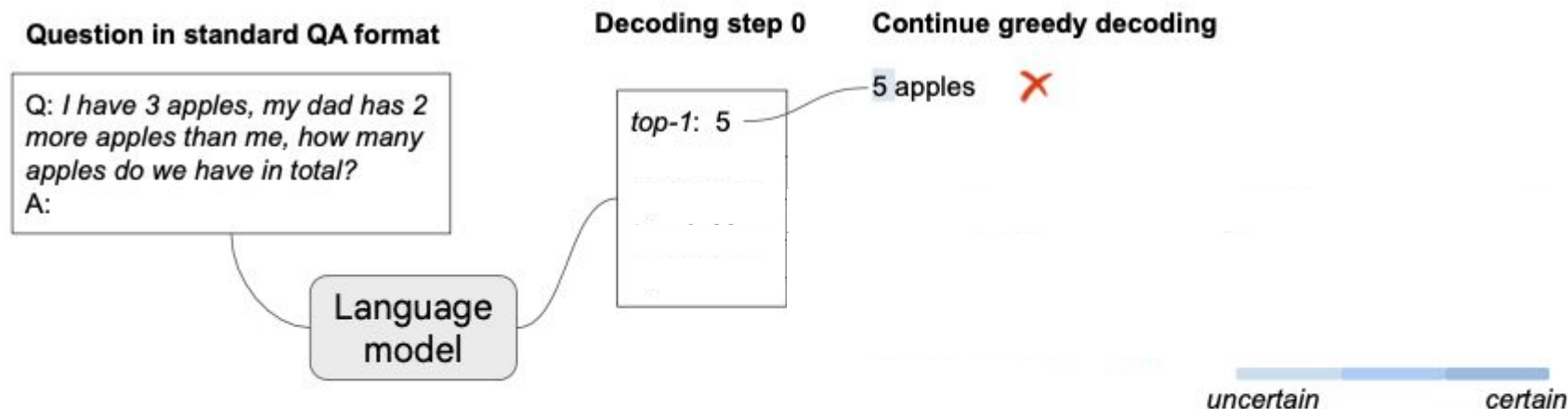


Figure 1: Illustration of CoT-decoding. Pre-trained LLMs are capable of inherent reasoning without prompting by considering alternative top- k tokens, rather than solely relying on the top-1 greedy decoding path. Moreover, these models tend to display higher confidence in decoding the final answer (indicated by a darker shaded color) when a CoT reasoning path is present.

Greedy Decoding vs Chain-of-Thought Decoding

		PaLM-2 Pre-trained				PaLM-2 Inst-tuned
		X-Small	Small	Medium	Large	Large
GSM8K	greedy	9.0	14.3	21.0	34.8	67.8
	CoT-decoding	17.7 (+8.7)	35.1 (+20.8)	39.7 (+18.7)	61.5 (+26.7)	81.3 (+13.5)
MultiArith	greedy	7.5	15.8	36.8	75.0	93.7
	CoT-decoding	34.8 (+27.3)	43.5 (+27.7)	52.5 (+15.7)	86.7 (+11.7)	98.7 (+5.0)

Key takeaways

Pre-trained LLMs, without further finetuning, has been **ready** for **step-by-step reasoning**, but we need a **non-greedy decoding** strategy to **elicit** it.

When a **step-by-step reasoning** path is **present**, **LLMs** have much higher **confidence** in **decoding** the final answer **than direct-answer decoding**.

Why "generate from the model" instead of "from humans"?

First principle in ML

Directly optimize what we want!

So what do we want to optimize here?

Optimize the **metric of measuring generation quality**.

Examples: correctness for solving math problems, BLUE score for machine translation, ...

All the rest is to compute gradients and do backpropagation!

So what do we want to optimize here?



Always keep in mind that LLMs are probabilistic models of generating next tokens. **They are not humans.**

What LLM does in decoding:

$$\arg \max \mathbb{P}(\text{reasoning path, final answer} | \text{problem})$$

One-step further

$$\begin{aligned} & \arg \max \mathbb{P}(\text{final answer} | \text{problem}) \\ = & \sum_{\text{reasoning path}} \mathbb{P}(\text{reasoning path, final answer} | \text{problem}) \end{aligned}$$

How to compute the sum then? Sampling!

CoT prompting greedily decodes the optimal reasoning path

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: Janet's ducks lay 16 eggs per day. She eats three for breakfast every morning and bakes muffins for her friends every day with four. She sells the remainder for \$2 per egg. How much does she make every day?

A:

Language
model

Greedy decode

This means she uses $3 + 4 = 7$ eggs every day. She sells the remainder for \$2 per egg, so in total she sells $7 * \$2 = \14 per day.
The answer is \$14.

Self-consistency: step 1 - prompt

Prompt with chain of thought

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: Janet's ducks lay 16 eggs per day. She eats three for breakfast every morning and bakes muffins for her friends every day with four. She sells the remainder for \$2 per egg. How much does she make every day?

A:

Language
model

Self-consistency: step 2 - sample decode

Prompt with chain of thought

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: Janet's ducks lay 16 eggs per day. She eats three for breakfast every morning and bakes muffins for her friends every day with four. She sells the remainder for \$2 per egg. How much does she make every day?

A:

Language model

Sample decode with diverse reasoning paths

She has $16 - 3 - 4 = 9$ eggs left. So she makes $\$2 * 9 = \18 per day.

The answer is \$18.

This means she uses $3 + 4 = 7$ eggs every day. So in total she sells $7 * \$2 = \14 per day.

The answer is \$14.

She eats 3 for breakfast, so she has $16 - 3 = 13$ left. Then she bakes muffins, so she has $13 - 4 = 9$ eggs left. So she has $9 \text{ eggs} * \$2 = \18 .

The answer is \$18.

Note the reasoning paths are optional, so they can be marginalized out

Self-consistency: step 3 - majority vote

Prompt with chain of thought

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: Janet's ducks lay 16 eggs per day. She eats three for breakfast every morning and bakes muffins for her friends every day with four. She sells the remainder for \$2 per egg. How much does she make every day?

A:

Language model

Sample decode with diverse reasoning paths

She has $16 - 3 - 4 = 9$ eggs left. So she makes $\$2 * 9 = \18 per day.

The answer is \$18.

This means she uses $3 + 4 = 7$ eggs every day. So in total she sells $7 * \$2 = \14 per day.

The answer is \$14.

She eats 3 for breakfast, so she has $16 - 3 = 13$ left. Then she bakes muffins, so she has $13 - 4 = 9$ eggs left. So she has $9 \text{ eggs} * \$2 = \18 .

The answer is \$18.

Majority vote on the answers

The answer is \$18.

Self-consistency

No fine-tuning, no human annotation, no additional modules like a verifier or a re-ranker

Performance gains across

Four LLMs with varying scales: UL2-20B, LaMDA-137B, PaLM-540B, GPT-3 (175B)

SoTA performance across various reasoning benchmarks

arithmetic, commonsense, and symbolic

How about free-from answers?

Universal Self-Consistency (USC)

**Ask LLMs to self-select the most
consistent answer**

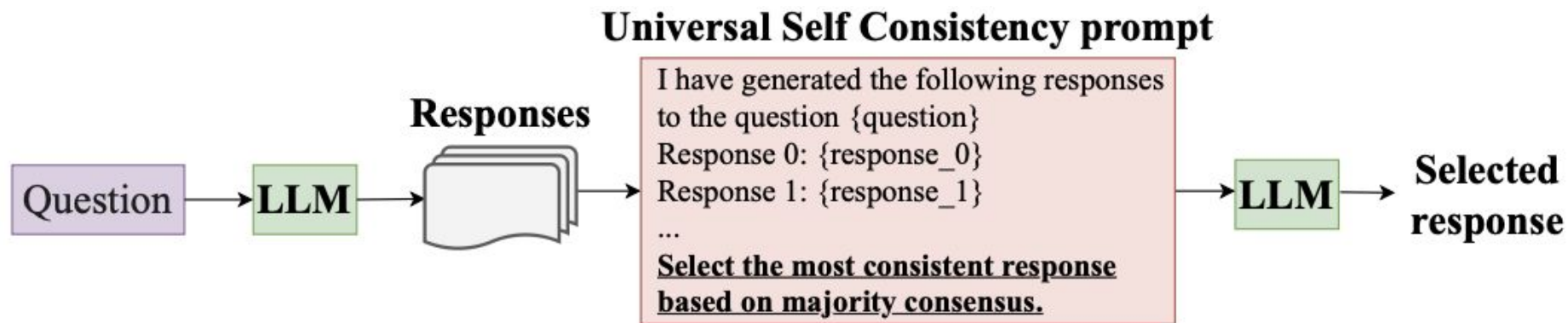


Figure 1: Overview of the Universal Self-Consistency workflow.

Limitations

LLMs Can Be Easily Distracted by Irrelevant Context

Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed Chi, Nathanael Schärli, and Denny Zhou. [Large Language Models Can Be Easily Distracted by Irrelevant Context](#). ICML 2023.

Adding irrelevant contexts to GSM8K leads to 20+ points performance drop

[Prompt] Lucy has \$65 in the bank. She made a \$15 deposit and then followed by a \$4 withdrawal. Maria's monthly rent is \$10. What is Lucy's bank balance?

[Response] Lucy's bank balance is $\$65 + \$15 - \$4 - \$10 = \$66$. The answer is \$66.

LLMs Cannot Self-Correct Reasoning Yet

Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, Denny Zhou. [Large Language Models Cannot Self-Correct Reasoning Yet](#). ICLR 2024.

Christina is planning a birthday party and needs .75 gift bags per invited guest, because $\frac{1}{4}$ of attendees don't show up. She invited 16 friends. Gift bags are \$2 each. How much will she spend?

... $0.75 * 12 = 9$ gift bags ... Christina will spend a total of $2 * 9 = \$8$... (Wrong)

Review your previous answer and find problems with your answer.

Upon reviewing my previous answer, I recognize a mistake. I incorrectly stated that Christina needs 0.75 gift bags per invited guest. The given information states that $\frac{1}{4}$ of the attendees won't show up...

Based on the problems you found, improve your answer.

... $16 * (\frac{3}{4}) = 12$ gift bags ... Christina will spend $12 * \$2 = \24 ... (Correct)

Terry eats 2 yogurts a day. They are currently on sale at 4 yogurts for \$5.00. How much does he spend on yogurt over 30 days?

... So, Terry spends \$75 on yogurt over 30 days. (Correct)

Review your previous answer and find problems with your answer.

Upon reviewing my previous answer, I realized that I made a mistake in calculating Terry's spending on yogurt over 30 days. I incorrectly stated that Terry spends \$2.50 per day for 30 days...

Based on the problems you found, improve your answer.

... the final answer is Terry spends \$37.5 on yogurt over 30 days. (Wrong)

Premise Order Matters in LLM Reasoning

Xinyun Chen, Ryan A Chi, Xuezhi Wang, Denny Zhou. [Premise Order Matters in Reasoning with Large Language Models](#). ICML 2024.

[Original GSM8K Problem] Thomas withdraws \$1000 in 20 dollar bills from the bank account. He loses 10 bills while getting home. After that, he uses half of the remaining bills to pay for a bill. Thomas then triples his money. He then converts all his bills to 5 dollar bills. How many 5 dollar bills does he have?

[Reordered GSM8K Problem] Thomas withdraws \$1000 in 20 dollar bills from the bank account. After getting home, he uses half of the remaining bills to pay for a bill. Thomas then triples his money. He then converts all his bills to 5 dollar bills. He loses 10 bills while getting home. How many 5 dollar bills does he have?

About 10 points drop on solving rates across all frontier LLMs

Summary

Summary

Generating intermediate steps improves LLMs performance

Training / finetuning / prompting with intermediate steps

Zero-shot, analogical reasoning, special decoding

Self-consistency greatly improves step-by-step reasoning

Limitations

irrelevant context, self-correction, premise order

Appendix

Tools and Applications

Tools & IDEs

- There are many tools, libraries, and platforms with different capabilities and functionalities
- Capabilities include:
 - Developing and experimenting with prompts
 - Evaluating prompts
 - Versioning and deploying prompts



PROMPTABLE

More tools here: <https://github.com/dair-ai/Prompt-Engineering-Guide#tools--libraries>

Coming next...

Augmentation

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